

X-RAY TUBE MECHANICAL ALIGNMENT

NOTE

For domestic shipments the x-ray tube is shipped installed; therefore, this procedure should not be necessary at installation. If you installed the tube, you must perform this procedure.

Introduction

This procedure explains how to mechanically align the X-ray tube so that the X-ray beam is centered on the detectors. This procedure must be done whenever an X-ray tube is replaced/installed. You will be working with the Gantry covers open while performing this procedure and will be initiating X-rays.

Tools Required

1. FSE tool kit
2. 3/8-inch drive socket wrench set

Materials Required

1. X-ray film – part number 97806

XRAY TUBE ALIGNMENT PROCEDURE

WARNING



SHOCK HAZARD. USE EXTREME CAUTION WHEN SERVICING THE GANTRY WITH THE COVERS OPEN. 480 VAC, 208 VAC, 120 VAC AS WELL AS 5VDC AND 15 VDC ARE EXPOSED IN MANY LOCATIONS. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.

1. Remove Gantry power. Refer to the [Power Up/Down procedures](#) in this manual.
2. Open and secure the upper Gantry cover. Remove the lower cover cone. Store the front cone out of the way. Do not lean the cone plastic inner edge of the cone against a wall as this might damage it.
3. Rotate the scan frame until the X-ray tube is at the 12 o'clock position. See Figure 66.

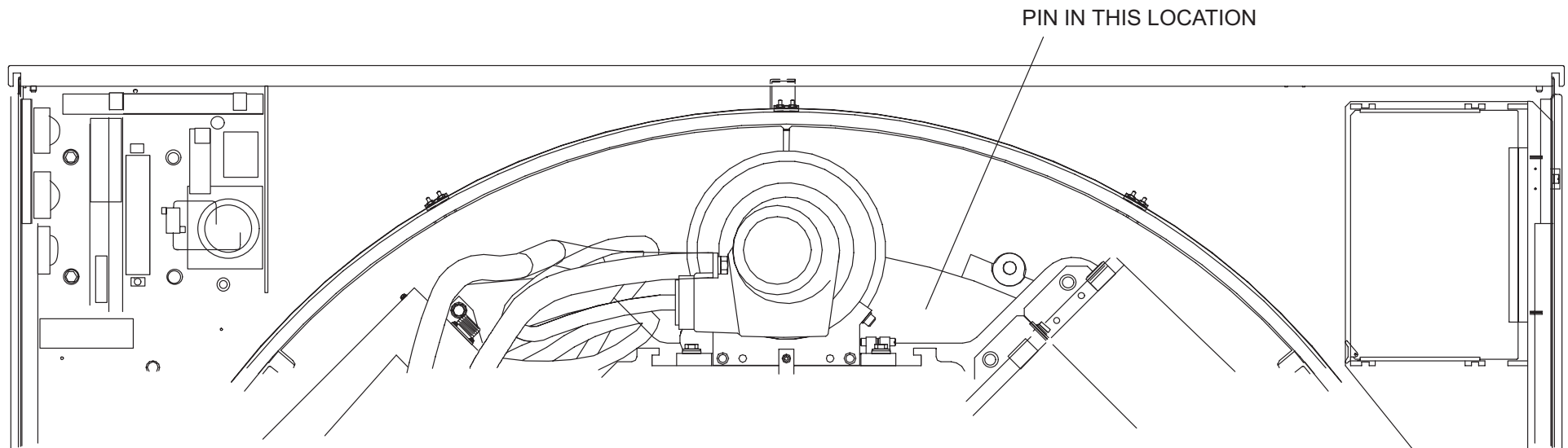


FIGURE 66 ROTATING FRAME WITH TUBE PINNED AT 12 O'CLOCK

4. Loosen the four tube cradle mounting bolts but do not remove them. Carefully turn the lateral adjustment screw until the hash mark on the cradle lines up with the hash mark on the base plate. See Figure 67.

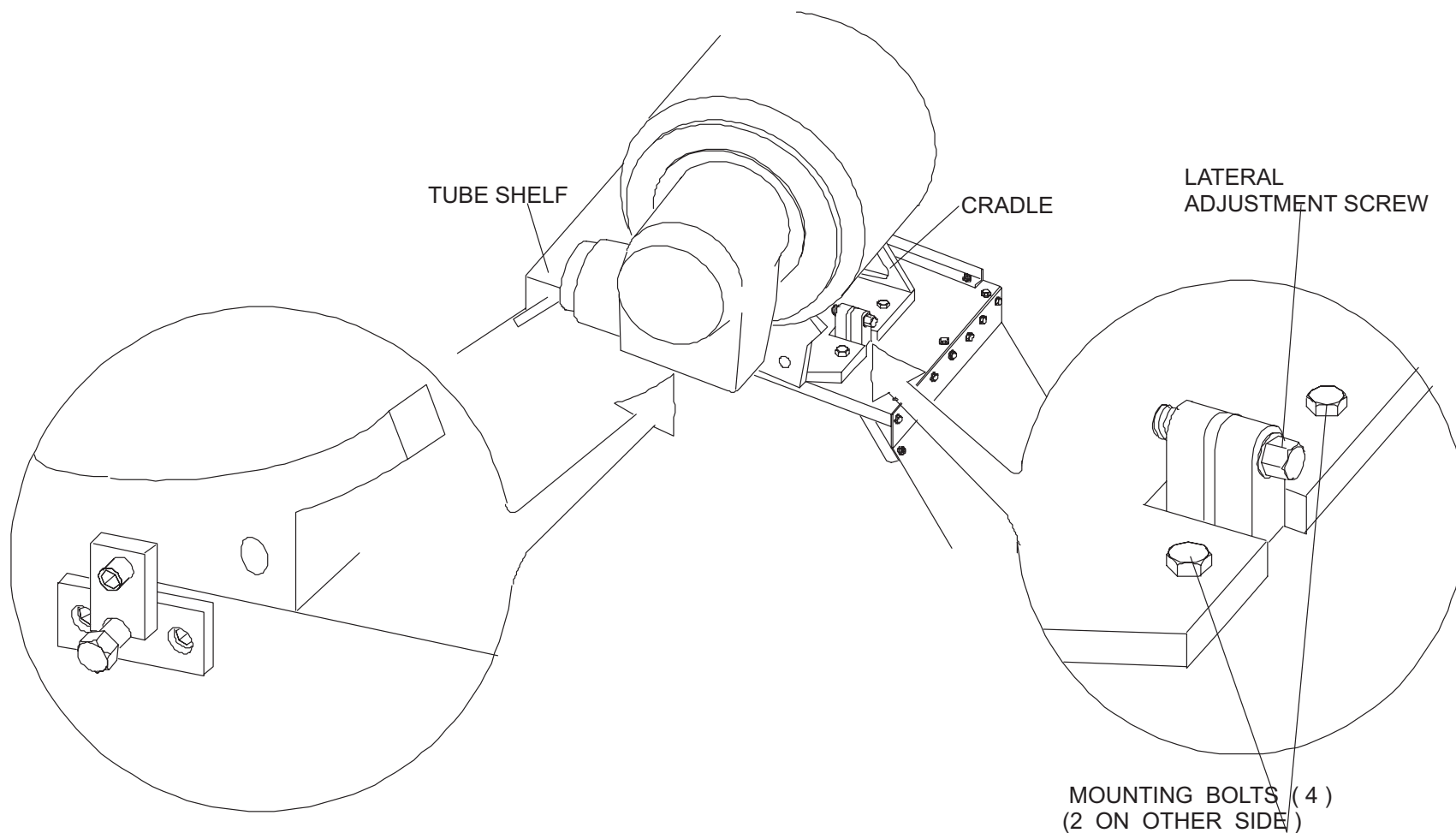


FIGURE 67 X-RAY TUBE ALIGNMENT

5. Tighten the four (4) cradle mounting bolts, in a cross pattern, to 15 ft-lbs.
6. Apply Gantry power. Turn the Display Tower and AcQImage Cabinet ON. Check that the main scan drives are off.
7. Do a Warm-up from the Scanner Utilities selection of the User Interface.
8. Disable the scan drives. When the upper front cover is open, the cover interlock switch automatically disables the scan drives. **DO NOT DEFEAT THE INTERLOCK SWITCH.**

9. Place a 4x5-inch film on the bottom of the inside of the Gantry aperture. Configure a reference marker in the orientation shown in Figure 68, centered in the x-ray field.

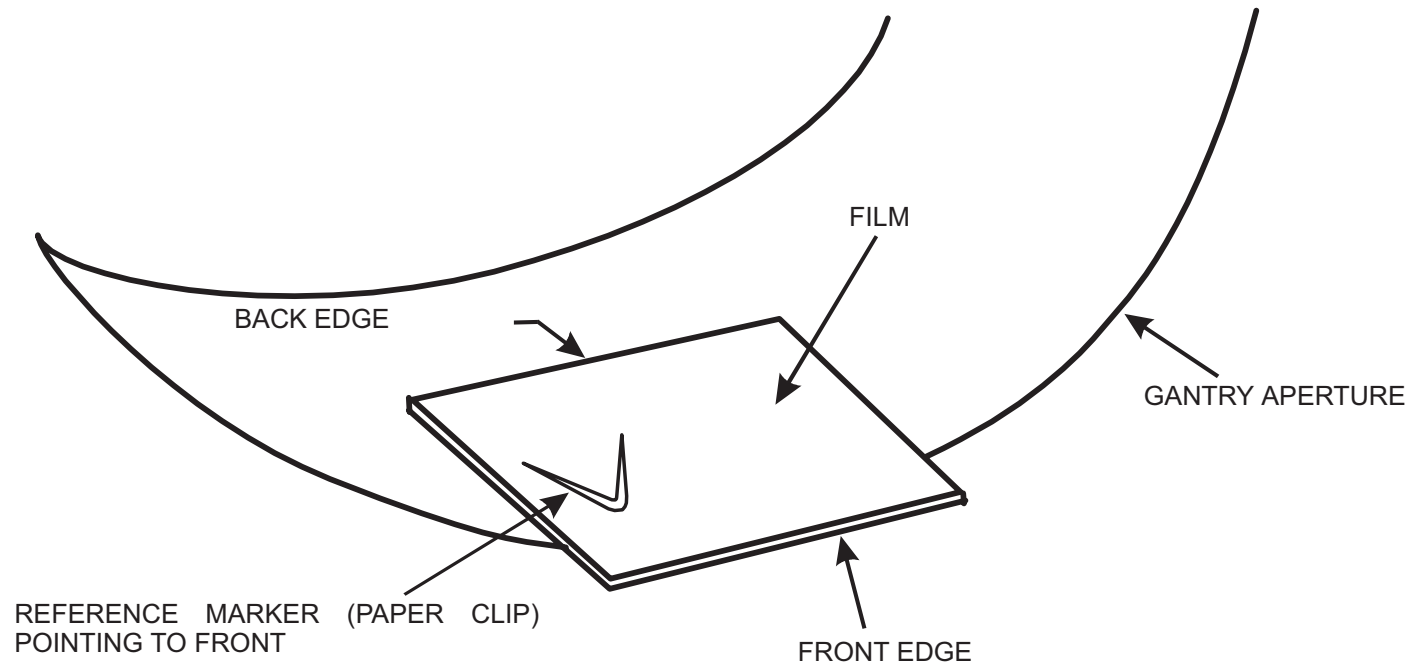


FIGURE 68 FILM AND REFERENCE MARKER

10. Turn scan drives on.
11. Position the X-ray tube at the 12 o'clock position (facing detector 1800).
12. Perform a stationary x-ray exposure (no scan frame motion). Refer to the new [MANEXP](#) procedure.

NOTE 1: The value for length of exposure is film speed dependent. Use an initial value of 200.

NOTE 2: The "tube position" option lets you use the rotate motor to position the x-ray tube if the scan frame drive is enabled. This number is the detector position where the tube will be located. Use the following table for examples:

MOVE TO DETECTOR	1800	600
..FACE DETECTOR ..	600	1800
Do Not Move Rotating Frame	-1	

13. Re-position the X-ray tube at the 6 o'clock position (facing detector = 600). Be careful to keep the film stationary.
14. Perform a stationary x-ray exposure (no scan frame motion) using [MANEXP](#) .
15. Note the film orientation with the reference object, so that when the film is developed, the image can be positioned correctly to make the centering measurements required.
16. Develop the film and inspect for a narrow, darker band superimposed on a wider, lighter band.

NOTE

The shadow of the X-ray beam on the film should show an even penumbra (fade off) on both sides to ensure mechanical centering.

17. Measure the distance from the front edge (see Figure 69) of the narrow, darker band to the front edge of the wide, lighter band (referred to as distance D1).
18. Measure the distance from the back edge of the narrow, darker band to the back edge of the wide, lighter band (referred to as distance D2).
19. Determine the distance between the band centers by the following equation:

$$\text{Distance between band centers (D)} = \frac{D1 - D2}{2}$$

NOTE

One full turn of the adjustment screw corresponds to a 0.025" (0.63 mm) movement of the X-ray tube.

20. Multiply the distance between band centers (D) by 0.417 to determine the distance the X-ray tube must move. Refer to the table below Figure 69) to determine which way to rotate the alignment screw.

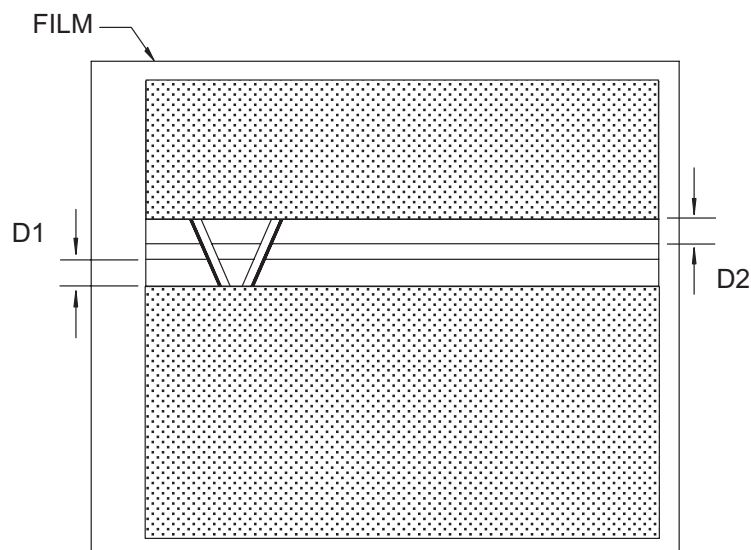


FIGURE 69 EXPOSED FILM

If the band is shifted to the FRONT,	rotate the screw CCW.
If the band is shifted to the REAR,	rotate the screw CW.

21. Loosen the four (4) tube cradle mounting bolts but do not remove them. See Figure 70.
22. Carefully turn the axial adjustment screw by hand, in the direction determined above, until you remove thread backlash; that is, any screw rotation causes movement of the tube base.
23. Look at and note the index position on the axial adjustment screw. Turn it the number of turns and in the direction indicated:

If the band is shifted to the FRONT,	rotate the screw CCW.
If the band is shifted to the REAR,	rotate the screw CW.

24. Tighten the four (4) cradle mounting bolts to 15 ft–lbs, in a cross pattern.
25. Repeat steps 10 through 19. Verify that the centers of the bands on the film are less than 0.25 mm apart.
26. Turn the Console OFF. Remove Gantry power. Refer to the [Power Up/Down](#) procedures in this manual.

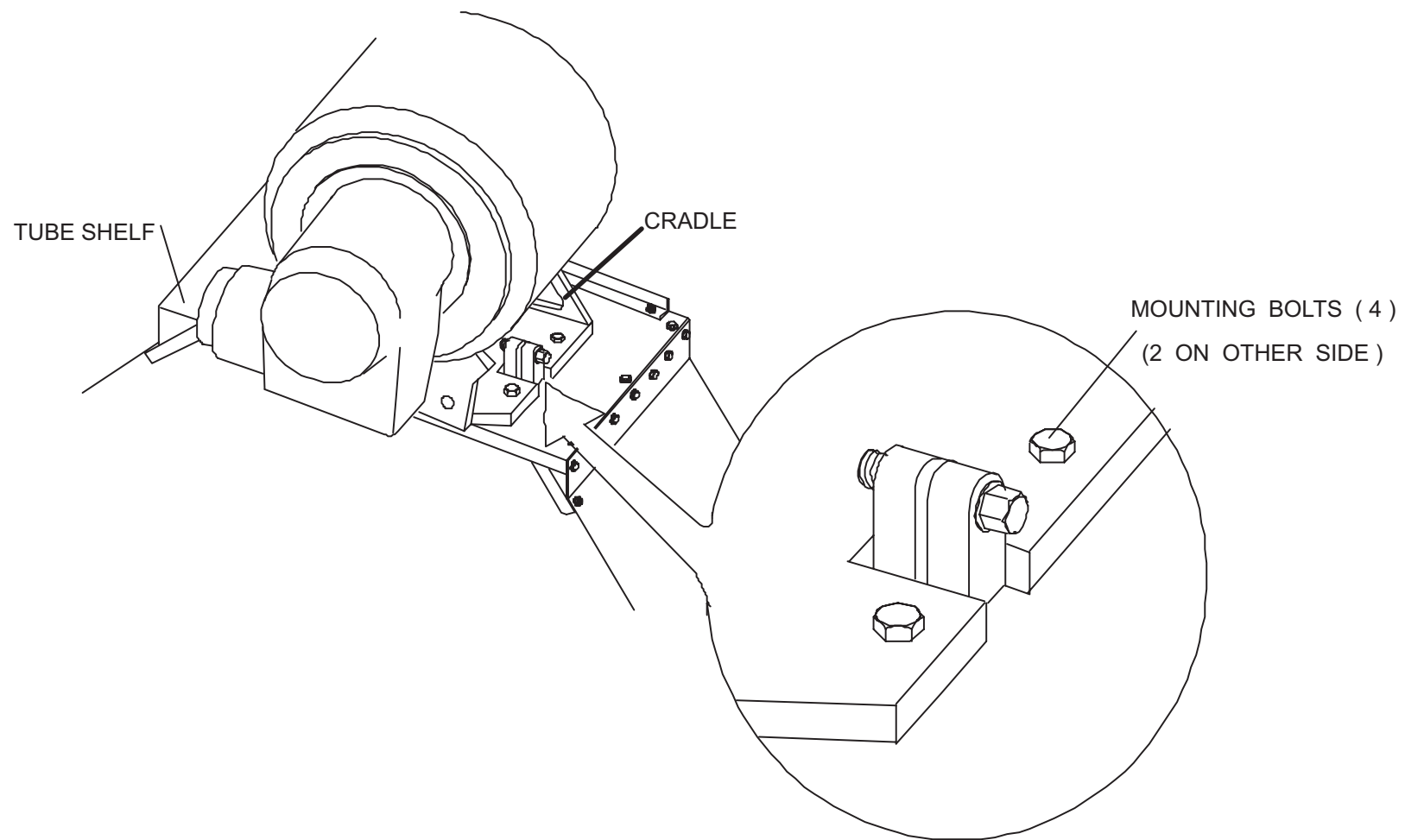


FIGURE 70 TYPICAL TUBE CRADLE ADJUSTMENT

COVERS

27. Gantry cover documentation can be found in the [Repair Manual](#).
28. Apply Gantry power. Refer to the Power Up/Down procedures in this manual. Allow 5–10 minutes for detector stabilization before continuing.
29. Run the Tube Centering procedure: first lateral centering (mechanical); then center calibration Refer to the [Auto-Calibration](#) manual.

30. If centering deviates by more than ± 2 ray sums, adjust the offset of the scan drive fine resolver using the Scan Drive Calibration procedure in the Calibration manual.
31. Run the CNA noise test (X ray noise test) and record the value obtained (for 50–60 Hz range). Refer to the [Auto-Calibration](#) manual.
32. Perform the full [Auto-Calibration](#) procedure.
33. Remove Gantry power. Refer to the Power Up/Down procedures in this manual.